

## Research Article

## A motion-region-based center frequency optimization method for Gabor wavelets in phase-based displacement measurement

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## ABSTRACT

Phase-based displacement measurement has attracted increasing attention in structural vibration monitoring because of its robustness to illumination variations and high subpixel accuracy. The Gabor wavelet is one of the earliest and most widely used phase extraction methods for phase-based displacement measurement. When Gabor wavelets are used for phase extraction, measurement performance is highly sensitive to the center frequency, while its selection in most previous studies relies on comparison with ground truth. To overcome this limitation, this study proposes a motion-region-based center frequency optimization method for Gabor wavelets in phase-based displacement measurement. Numerical experiments on numerical videos with different target motion characteristics show that the optimal center frequency varies systematically with these characteristics. In particular, the results reveal a clear relationship between the optimal center frequency and the target motion region, with larger motion regions generally requiring lower center frequencies for accurate displacement measurement. Based on this finding and the physical interpretation of the Gabor wavelet, a physics-inspired relationship between the optimal center frequency and the motion region is established, and an iterative center frequency optimization method is developed accordingly. The proposed optimization method is validated through a laboratory shaker test and an outdoor seismic response measurement of a cold-formed steel wall structure.

## 1. Introduction

In structural health monitoring, structural vibration displacement response is a key parameter for evaluating structural performance. It provides critical information for identifying variations in natural frequencies, stiffness, and modal properties, thereby enabling a comprehensive assessment of structural health status (Dong & Catbas, 2021). Throughout their service life, civil structures are subjected to various external loads, including traffic, seismic, and wind loads. Therefore, monitoring structural responses under these loading conditions in both laboratory and field environments is essential for performance evaluation and safety assurance (Dong & Catbas, 2021).

With advancements in camera technologies and computer vision techniques, vision-based methods hold significant potential for displacement measurement in the field of structural health monitoring. Compared to traditional methods for contact and non-contact displacement measurements, such as the employment of an accelerometer and laser displacement sensor, vision-based methods offer

numerous advantages, including low installation costs, affordability of devices, remote sensing capabilities, high-resolution measurements, and the absence of mass loading effects (Feng & Feng, 2018; Xu & Brownjohn, 2018; Chen et al., 2018; Gomez et al., 2018; Saxena & Seksen, 1989).

However, the accuracy and robustness of the vision-based method can be significantly influenced by the algorithms used for video processing (Feng & Feng, 2018; Xu & Brownjohn, 2018). Therefore, this study mainly focuses on displacement extraction algorithms for vision-based measurement. In general, these algorithms can be classified into two categories: intensity-based methods and phase-based methods (Miao et al., 2023a).

Intensity-based displacement measurement methods mainly include template matching, feature point matching, conventional optical flow, and recent deep learning-based optical flow approaches (Feng & Feng, 2018). Template matching tracks the motion of a selected region by searching for the most similar image patch in consecutive frames and can achieve high precision with subpixel refinement (Zhang et al.,

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2016). Owing to its simple implementation, it has been widely used in vision-based displacement monitoring, although its performance may degrade under illumination variation, occlusion, or target deformation (Dong & Catbas, 2021). Feature point matching detects and tracks salient key points and can provide reliable matching for regions with distinctive features; however, its effectiveness depends on the availability of such features and is therefore limited in low-texture scenes (Yoon et al., 2016). Compared with matching-based methods, intensity-based optical flow enables full-field displacement measurement with subpixel accuracy (Horn & Schunck, 1981). The Lucas–Kanade (LK) method improves the stability of optical flow estimation by assuming consistent motion within a local neighborhood, but its accuracy may still decrease in the presence of large displacement, noise, or illumination changes (Lucas & Kanade, 1981). More recently, deep learning has significantly advanced optical flow estimation. A representative method is the Recurrent All-Pairs Field Transforms (RAFT) model, which builds dense correspondences between two frames and iteratively refines the flow field, enabling accurate recovery of both large motions and fine details (Teed & Deng, 2020). Overall, intensity-based methods remain sensitive to imaging conditions and generally require stable illumination, sufficient surface texture, and a clear background.

The aforementioned intensity-based methods are highly sensitive to illumination changes, requiring careful control of lighting conditions and the preparation of markers and backgrounds, which often involve extensive preparatory work. As a result, phase-based displacement measurement methods have gained increasing attention in recent years due to their robustness to photometric variations and high subpixel accuracy (Fleet & Jepson, 1990; Barron et al., 1992; Diamond et al., 2017; Collier & Dare, 2022; Cai et al., 2023; Liu et al., 2024; Li et al., 2024).

Typically, phase-based displacement measurement consists of two primary stages: phase extraction and the calculation of phase-based optical flow (Fleet & Jepson, 1990). Among them, phase extraction is the more critical step, because it provides the phase information required for subsequent optical flow computation. In addition, several parameters in phase extraction methods can affect displacement measurement performance (Diamond et al., 2017; Miao et al., 2022; Wang et al., 2023; Ai et al., 2025). The choice of phase generation technique is relatively flexible, and several well-established traditional phase extraction methods have been widely employed in this field (Fleet & Jepson, 1990; Diamond et al., 2017; Miao et al., 2022; Portilla & Simoncelli, 2000).

Among these, the Gabor wavelet transform is one of the earliest and most widely used phase extraction methods for phase-based displacement measurement. In 1990, (Fleet & Jepson, 1990) first elucidated the close relationship between the phase and motion of a pixel, and based on this foundational insight, they employed a spatial-temporal Gabor wavelet to generate phase for phase-based pixel velocity estimation. (Gautama & Van Hulle, 2002) adopted a bank of the 2D Gabor wavelets to generate the phase information of image sequence for pixel component velocity estimation and computed the full velocity of pixels based on the component velocities. The Gabor wavelet was used for modal identification and motion magnification of a wind blade in a camera-based laboratory experiment (Sarraf et al., 2018). Miao et al. (2022) utilized the Gabor wavelet transform in conjunction with phase-based optical flow as an image processing algorithm to accurately and robustly measure the displacement of a frame structure in a laboratory experiment. To circumvent the issues of direct current (DC) components and negative frequencies that arise with high bandwidths in Gabor wavelet, (Wang et al., 2023) employed a log Gabor wavelet for enhanced displacement measurement. (Zhang et al., 2024) employed the improved particle swarm optimization (IPSO) algorithm to optimize the parameters of the log-Gabor wavelet for phase-based displacement measurement.

Previous studies have shown that the center frequency of the Gabor

wavelet significantly affects the displacement measurement performance of phase-based methods (Miao et al., 2022; Wang et al., 2023; Zhang et al., 2024). In most of these studies, the center frequency was optimized by comparing the measurement results with ground truth motion (Wang et al., 2023; Zhang et al., 2024). However, in real-world applications, ground truth displacement data are not always available. A prior study (Ai et al., 2025) on phase-based displacement measurement using the complex steerable pyramid proposed a ground-truth-free optimization strategy based on an empirical relationship between the optimal scale level and motion characteristics, but this relationship was developed under a specific video resolution, making it resolution-dependent. Therefore, it is crucial to develop a more general optimization method for the center frequency of the Gabor wavelet that does not rely on ground truth or a specific video resolution.

Considering that the algorithm is applied to motion videos, ground truth is no longer needed if the center frequency can be optimized based on the motion characteristics of the video. Motivated by this idea, this study proposes a center frequency optimization method based on target motion characteristics for the Gabor wavelet in phase-based displacement measurement, enabling center frequency optimization without reliance on ground truth. In particular, the relationship between the optimal center frequency, determined by comparison with ground truth, and target motion characteristics is systematically investigated. Virtual videos with different motion characteristics are generated to explore this relationship. Based on the identified relationship, a ground-truth-free center frequency optimization method is established and subsequently validated through both laboratory and outdoor experiments.

The main contributions of this study are summarized as follows. First, it identifies a previously unexplored relationship between the optimal center frequency of the Gabor wavelet and the target motion region, which is determined by the motion range and target size. Rather than treating center frequency selection as a case-dependent tuning problem, this relationship provides a physics-inspired and interpretable foundation for parameter optimization. Second, based on this relationship, a practical iterative optimization method is developed that determines the optimal center frequency directly from video data without relying on ground truth displacement information. Third, the proposed method is validated through both laboratory and outdoor shaking table experiments, demonstrating improved measurement accuracy and strong practical potential for structural vibration monitoring.

The structure of this paper is organized as follows: Section 2 introduces the methodology of phase-based displacement measurement using the Gabor wavelet, including the Gabor wavelet transform, its application in phase-based optical flow, and parameter selection for the Gabor wavelet. Section 3 presents a center frequency optimization method for Gabor wavelets in phase-based displacement measurement. Specifically, numerical experiments based on numerical videos with different motion characteristics are conducted to investigate the optimal center frequency under different motion conditions, the relationship between the optimal center frequency and motion region is established, and a ground-truth-free parameter optimization method is proposed accordingly. Section 4 validates the proposed method through experimental studies, including a shaker test on a fixed column and the measurement of seismic response of a cold-formed steel wall structure.

## 2. Methodology

### 2.1. Gabor wavelet transform

Developed prior to the discrete wavelet method, the Gabor wavelet transform, also known as the Morlet wavelet transform, is a type of continuous wavelet transform that offers an overcomplete representation (Simpson, 1993).

The 2D Gabor wavelet transform, primarily performed by convolving the original image with a 2D Gabor wavelet, is renowned for its ability to capture directional information within an image. This filter comprises

both even and odd pairs, closely resembling the receptive fields of the human visual cortex (Daugman, 1985; Fogel & Sagi, 1989).

Consisting of complex sinusoids and a Gaussian envelope (Movellan, 2002), the 2D Gabor wavelet is illustrated below.

$$G(x, y; \theta, \gamma, f, \sigma, \psi) = \exp\left(-\frac{x'^2 + \gamma^2 y'^2}{2\sigma^2}\right) \exp(j(2\pi f x') + \psi) \quad (1)$$

$$x' = x \cos \theta + y \sin \theta \text{ and } y' = -x \sin \theta + y \cos \theta \quad (2)$$

where  $\theta$  is the direction,  $\gamma$  is the aspect ratio in the spatial domain,  $\psi$  is the phase offset and  $f$  is the center frequency of the Gabor wavelet.  $\sigma$  is the standard deviation of the Gaussian envelope that decides the size of the Gabor wavelet in the spatial domain.

The frequency bandwidth  $\beta$  of the 2D Gabor wavelet is shown in Eq. (3), which is determined by  $f\sigma$ .

$$\beta = \frac{\ln\left(\frac{2\pi f \sigma + \sqrt{2\ln(2)}}{2\pi f \sigma - \sqrt{2\ln(2)}}\right)}{\ln(2)} \quad (3)$$

Several parameters can be adjusted in the Gabor wavelet, including the standard deviation of the Gaussian envelope, bandwidth, center frequency, and the direction of the Gabor transform.

By convolving the 2D Gabor wavelet with the image signal, a complex result  $R(x, y)$  is obtained,  $A(x, y)$  is the amplitude of the complex result, and  $\phi(x, y)$  is the phase of the complex result, as shown in Eq. (4).

$$R(x, y) = G \otimes I = A(x, y)e^{j\phi(x, y)} \quad (4)$$

## 2.2. The application of the Gabor wavelet transform in phase-based optical flow

In 1990, (Fleet & Jepson, 1990) introduced the phase constancy assumption for phase-based optical flow displacement measurement, akin to the brightness constancy assumption in traditional intensity-based optical flow. Unlike intensity information, which contains bidirectional motion data and contributes to the aperture problem in intensity-based optical flow, phase information is inherently unidirectional. Therefore, utilizing phase information from a specific direction to compute motion, based on the phase constancy assumption, is a conventional way in phase-based optical flow that effectively circumvents the aperture problem associated with intensity-based techniques.

To obtain the full velocity using the phase-based optical flow method, it is necessary to compute the velocity in various directions using the phase information of different orientations. Once the velocities in different directions are obtained, a mathematical approach, such as the least squares method, can be used to address the overdetermined problem and determine the full pixel velocity (Miao et al., 2023a; Gautama & Van Hulle, 2002).

The convolution of the Gabor wavelet with an image yields a complex number, and the phase of the image can be computed from the phase angle of this complex result. Once the phase information is obtained, the phase constancy assumption in Fleet et al.'s work can be utilized to compute the velocity of each pixel (Fleet & Jepson, 1990). Subsequently, the displacement of pixels can be calculated from frame to frame. The process of phase-based optical flow displacement measurement of video using the Gabor wavelet transform is described below.

First, the video signal is convolved with a 2D Gabor wavelet, having a sequence of complex results  $R(x, y, t)$ , as shown in Eq. (5).

$$R(x, y, t) = G \otimes I(x, y, t) = A(x, y, t)e^{j\phi(x, y, t)} \quad (5)$$

Then, the phase information  $\phi(x, y, t)$  is computed from the complex result via the phase angle method.

With the assumption of phase constant, the derivation with respect to time can be done on each side of Eq. (6) to obtain Eq. (7). In Eq. (7),  $\phi_x$  represents the spatial phase gradient with respect to the horizontal

direction of the frame,  $\phi_y$  denotes the vertical spatial phase gradient, and  $\phi_t$  is the temporal phase gradient. Additionally,  $v_x$  and  $v_y$  correspond to the velocities of the pixel in the horizontal and vertical directions, respectively.

$$\phi(x, y, t) = c \quad (6)$$

$$\phi_x v_x + \phi_y v_y + \phi_t = 0 \quad (7)$$

Projecting the Eq. (7) to the direction of Gabor wavelet results in Eq. (8).

$$\|\vec{\phi_c}\| \|\vec{v}\| \cos \alpha + \phi_t = 0 \quad (8)$$

The full velocity of pixel  $\vec{v}$  is a vector in any direction.  $\vec{\phi_c}$  is the phase gradient. The vectors  $\vec{\phi_c}$  and  $\vec{v}$  form an angle  $\alpha$  between them.

Then, the component velocity  $\vec{v_c}$  in the direction of the Gabor wavelet can be computed based on Eq. (9). In Eq. (9),  $\vec{\phi_c}^n$  is the unit vector of the phase gradient.

$$\begin{aligned} \vec{v_c} &= \vec{\phi_c}^n \|\vec{v}\| \cos \alpha = \vec{\phi_c}^n \left( -\frac{\phi_t}{\|\vec{\phi_c}\|} \right) = \frac{(\phi_x, \phi_y)}{\|\vec{\phi_c}\|} \left( -\frac{\phi_t}{\|\vec{\phi_c}\|} \right) \\ &= -\frac{\phi_t}{(\phi_x^2 + \phi_y^2)} (\phi_x, \phi_y) \end{aligned} \quad (9)$$

With the velocity of pixels known, the motion of pixels can be computed based on the time interval and velocity, as shown in Eq. (10).

$$\vec{d_c} = \vec{v_c} \Delta t \quad (10)$$

## 2.3. The parameter selection for the Gabor wavelet

As previously mentioned, four parameters of the Gabor wavelet require specification: the standard deviation of the Gaussian envelope, the center frequency, the bandwidth, and the direction of the Gabor wavelet. The direction of the Gabor wavelet is usually aligned with the direction intended to extract displacement in video.

The selection of the Gabor wavelet's bandwidth requires a careful balance between spatial and frequency localization abilities (Daugman, 1985). Spatial localization, determined by the standard deviation of the Gaussian envelope or the size of the Gabor wavelet in the spatial domain, quantifies the accuracy with which the filter can pinpoint the location of a target in each frame. Frequency localization, on the other hand, reflects the filter's precision in isolating specific frequency components from an image, thus minimizing the impact of other frequency components (noise).

The product of the standard deviation of the Gaussian envelope and center frequency determines the bandwidth of the Gabor wavelet (Movellan, 2002; Miao et al., 2023b; Liu et al., 2022). From Eq. (11), it is evident that with a fixed center frequency  $f$ , the standard deviation  $\sigma$  and bandwidth  $\beta$  are inversely proportional. A smaller bandwidth enhances frequency localization ability, while a smaller standard deviation improves spatial localization. However, these two properties are inversely related, necessitating a trade-off.

$$2\pi f \sigma = \sqrt{2\ln(2)} \frac{2^\beta + 1}{2^\beta - 1} \quad (11)$$

Considering the need to balance spatial and frequency localization abilities effectively, we opt for a bandwidth of 1, as adopted in previous studies (Fleet & Jepson, 1990; Liu et al., 2022).

In the field of image feature detection, the frequency of an object often correlates with its characteristics. For example, thin objects such as edges typically exhibit high frequencies, while wider objects like surfaces display low frequencies (Morrone et al., 1986). Consequently, the selection of the center frequency for a Gabor wavelet is typically guided by the types of targets intended to be analyzed from the convolution

results with the image. For instance, to focus on high-frequency targets such as edges, the center frequency should be set to a high value.

However, in the field of displacement measurement, it remains unclear whether the motion range and target size correspond to a specific center frequency of the Gabor wavelet to capture the motion optimally (Miao et al., 2022; Zhang et al., 2024). Therefore, Section 3.1.2 investigates the motion measurement performance of the Gabor wavelet method under various center frequencies, motion amplitudes, and target sizes.

### 3. A center frequency optimization method for Gabor wavelets in phase-based displacement measurement

In this section, numerical videos with different motion characteristics, including various motion ranges and target sizes, are generated and measured using Gabor wavelets with different center frequencies in a phase-based displacement measurement framework. The variation of the optimal center frequency with respect to the target's motion characteristics is then investigated, and a physics-inspired mathematical relationship between them is established. Based on this relationship, a parameter optimization method that is independent of ground truth is proposed.

#### 3.1. Numerical experiment

##### 3.1.1. Numerical videos

The numerical video features an 8-bit depth, 720P resolution, at 50 frames per second (FPS), with a duration of 3 s. The motion target is a white circle ( $I = 255$ ) with diameters of 0.012, 0.024, 0.048, 0.096, 0.120, and 0.240 as fractions of the frame width (1280 pixels). The diameter-to-frame width ratio can be deemed as the target size ratio. The background is black ( $I = 0$ ). Initially placed at the center of the frame, the target exhibits free vibration in the horizontal direction with a damping coefficient  $c$  of 0.25 and natural frequency  $\omega_n$  of  $4\pi$ , as depicted in Eq. (12). The first frames of some cases in numerical motion videos are shown in Fig. 1.

$$x = Ae^{-ct} \sin(\omega_n t) \quad (12)$$

The amplitudes of motion,  $A$ , are defined as multiples of the frame width (1280 pixels), with values of 0.004, 0.008, 0.040, 0.080, 0.100, 0.150, 0.200, and 0.250. Given the minimal motion amplitude of approximately 5.12 pixels, calculated as 0.004 times 1280, the video frame is upsampled by a factor of 10 to generate the motion accurately. Bicubic interpolation is subsequently used to restore the original 720P resolution of the frame, achieving a motion resolution of 0.1 pixels. To more closely mimic real-world conditions, Gaussian noise with a standard deviation of 2 pixels is added to the video frames.

##### 3.1.2. The optimal center frequency of the Gabor wavelet

Twenty-five Gabor wavelets with center frequencies of 0.00075, 0.001, 0.00125, 0.0015, 0.00175, 0.002, 0.0025, 0.003, 0.0035, 0.004,

0.0045, 0.005, 0.0055, 0.006, 0.008, 0.01, 0.015, 0.02, 0.03, 0.05, 0.07, 0.1, 0.15, 0.2, and 0.3 are applied to each numerical motion video for phase-based displacement measurement. Those center frequencies are selected to fully cover the range of component sizes that may appear in a 720P video frame, ranging from  $1/0.3 = 3.33$  pixels to  $1/0.00075 = 1333.33$  pixels. As stated in Section 2.3, the bandwidth of the Gabor wavelet is 1, and the direction of the Gabor wavelet is horizontal. The point of interest (POI) is set at the center of the frame for all displacement measurement tasks.

Subsequently, the MAE is calculated for each case to determine the optimal center frequency of the Gabor wavelet. The optimal center frequency is selected based on the measurement that yields the minimum MAE, which most closely aligns with the ground truth. As demonstrated in Fig. 2, the phase-based optical flow based on the Gabor wavelet provides an accurate displacement measurement.

As shown in Table 1, for most cases, the measurement performance from the Gabor wavelet with optimal center frequency is good, producing accurate displacement measurement results. However, in scenarios involving small targets and large motion amplitudes, the performance of the displacement measurement is suboptimal. Specifically, for the case with a target size ratio of 0.012 and a motion amplitude ratio of 0.25, despite the considerable motion amplitude, it results in an MAE of 16.12 pixels, indicating a relatively large measurement error. This issue primarily arises from the extensive focus region of the Gabor wavelet in the spatial domain, which corresponds to a low center frequency (0.00175) in the frequency domain and a broad Gaussian envelope in the spatial domain. Enlarging the focus region to capture large motion leads to increased inclusion of noise within this

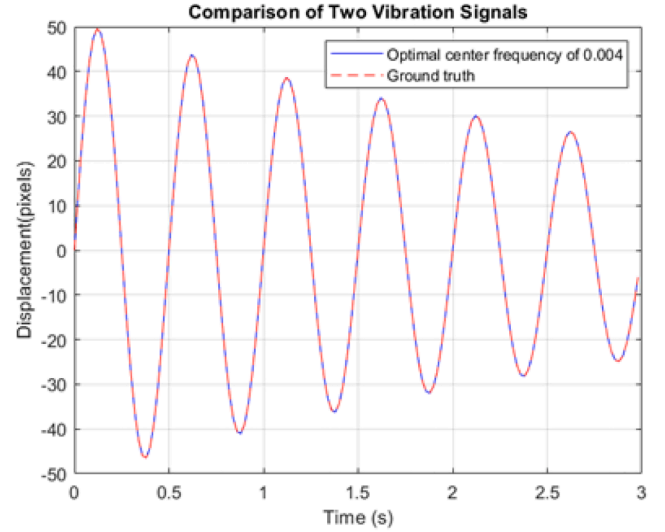
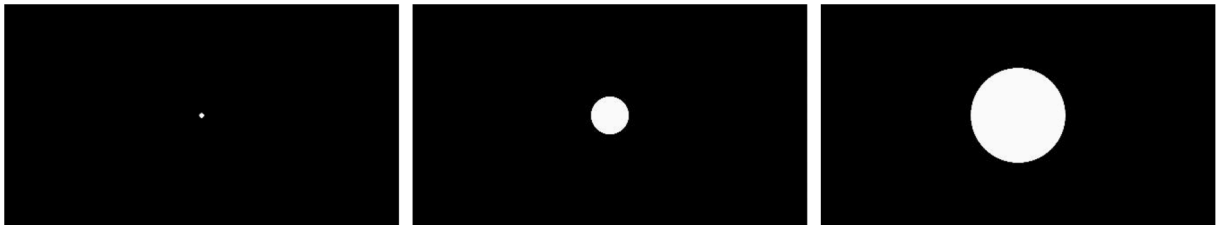


Fig. 2. Displacement measurement result using the Gabor wavelet with optimal center frequency (motion amplitude ratio = 0.040, target size ratio = 0.120).



(a) Target size ratio = 0.012

(b) Target size ratio = 0.096

(c) Target size ratio = 0.240

Fig. 1. First frames of numerical motion videos with different target size ratios.

**Table 1**

Optimal center frequency and MAE of Gabor wavelet method under different motion amplitude and target size of numerical videos.

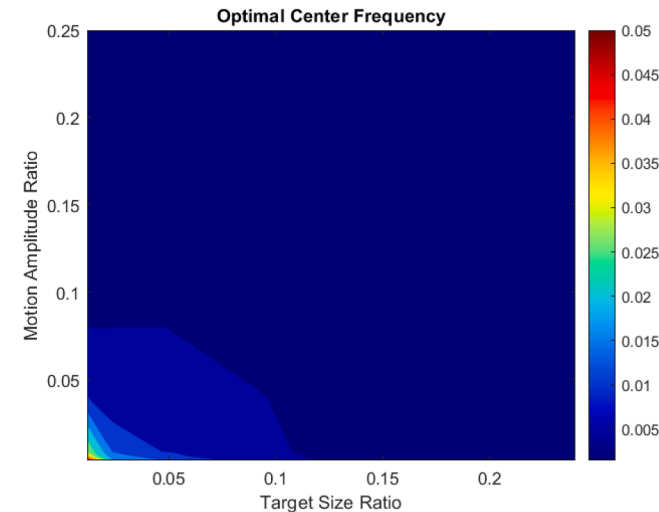
| Motion amplitude ratio | Target size ratio | Optimal center frequency | MAE (pixels) | Motion amplitude ratio | Target size ratio | Optimal center frequency | MAE (pixels) |
|------------------------|-------------------|--------------------------|--------------|------------------------|-------------------|--------------------------|--------------|
| 0.004                  | 0.012             | 0.0500                   | 0.03268      | 0.100                  | 0.012             | 0.00400                  | 0.77812      |
| 0.004                  | 0.024             | 0.0200                   | 0.03439      | 0.100                  | 0.024             | 0.00400                  | 0.20708      |
| 0.004                  | 0.048             | 0.0150                   | 0.01698      | 0.100                  | 0.048             | 0.00400                  | 0.08210      |
| 0.004                  | 0.096             | 0.0060                   | 0.01274      | 0.100                  | 0.096             | 0.00400                  | 0.02379      |
| 0.004                  | 0.120             | 0.0050                   | 0.01268      | 0.100                  | 0.120             | 0.00300                  | 0.02933      |
| 0.004                  | 0.240             | 0.0030                   | 0.01889      | 0.100                  | 0.240             | 0.00175                  | 0.02213      |
| 0.008                  | 0.012             | 0.0300                   | 0.03460      | 0.150                  | 0.012             | 0.00250                  | 2.04629      |
| 0.008                  | 0.024             | 0.0150                   | 0.04320      | 0.150                  | 0.024             | 0.00250                  | 0.63154      |
| 0.008                  | 0.048             | 0.0100                   | 0.01882      | 0.150                  | 0.048             | 0.00250                  | 0.17382      |
| 0.008                  | 0.096             | 0.0060                   | 0.01260      | 0.150                  | 0.096             | 0.00250                  | 0.04894      |
| 0.008                  | 0.120             | 0.0040                   | 0.01376      | 0.150                  | 0.120             | 0.00250                  | 0.05589      |
| 0.008                  | 0.240             | 0.0030                   | 0.01757      | 0.150                  | 0.240             | 0.00175                  | 0.02704      |
| 0.040                  | 0.012             | 0.0100                   | 0.09769      | 0.200                  | 0.012             | 0.00200                  | 4.34809      |
| 0.040                  | 0.024             | 0.0080                   | 0.08268      | 0.200                  | 0.024             | 0.00200                  | 0.91248      |
| 0.040                  | 0.048             | 0.0060                   | 0.02538      | 0.200                  | 0.048             | 0.00200                  | 0.27413      |
| 0.040                  | 0.096             | 0.0050                   | 0.01371      | 0.200                  | 0.096             | 0.00175                  | 0.10930      |
| 0.040                  | 0.120             | 0.0040                   | 0.01324      | 0.200                  | 0.120             | 0.00175                  | 0.05443      |
| 0.040                  | 0.240             | 0.0025                   | 0.01826      | 0.200                  | 0.240             | 0.00175                  | 0.04891      |
| 0.080                  | 0.012             | 0.0050                   | 0.39577      | 0.250                  | 0.012             | 0.00150                  | 16.12170     |
| 0.080                  | 0.024             | 0.0050                   | 0.12346      | 0.250                  | 0.024             | 0.00150                  | 3.92554      |
| 0.080                  | 0.048             | 0.0050                   | 0.04863      | 0.250                  | 0.048             | 0.00150                  | 1.06296      |
| 0.080                  | 0.096             | 0.0045                   | 0.03016      | 0.250                  | 0.096             | 0.00150                  | 0.19623      |
| 0.080                  | 0.120             | 0.0030                   | 0.01893      | 0.250                  | 0.120             | 0.00150                  | 0.13724      |
| 0.080                  | 0.240             | 0.0020                   | 0.02244      | 0.250                  | 0.240             | 0.00150                  | 0.09324      |

area, negatively impacting measurement performance. On the other hand, smaller targets, which resemble noise and represent minor components within the frame, are more susceptible to misidentification as noise, thereby further compromising the accuracy of measurements.

A contour map of the optimal center frequency of the Gabor wavelet concerning the motion amplitude and target size is plotted in Fig. 3, based on the data in Table 1. As demonstrated in Fig. 3, the optimal center frequency of the Gabor wavelet decreases with increasing motion amplitude and target size. Similar to image feature detection, where low center frequencies correspond to larger components such as surfaces, the Gabor wavelet method also requires low center frequencies to effectively accommodate large motion amplitudes and target sizes, to achieve optimal measurement performance.

### 3.2. The relationship between optimal center frequency and motion region

From the physical perspective of the center frequency in the Gabor wavelet, it is associated with the focus frequency of the Gabor wavelet in



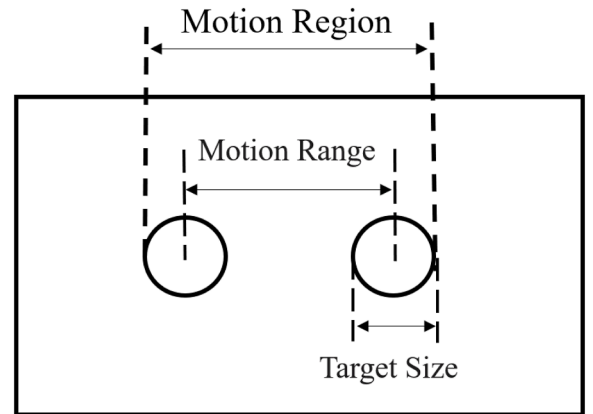
**Fig. 3.** Optimal center frequency contour based on motion amplitude ratio and target size ratio.

the frame (Movellan, 2002). The focus frequency is directly related to specific scale components in the frame, expressed in units of pixels (Fogel & Sagi, 1989; Movellan, 2002). The scale component corresponding to a given focus frequency defines the focus region of the Gabor wavelet in the frame. Therefore, a mathematical relationship exists between the center frequency and the focus region, as illustrated in Eq. (13).

$$\text{center frequency} = \frac{1}{\text{focus region}} \quad (13)$$

The target size and motion range can be jointly defined as the motion region, as illustrated in Fig. 4 and Eq. (14). If the focus region is smaller than the motion region, the motion measurement will fail because the target motion beyond the focus region cannot be properly localized. On the other hand, if the focus region is much larger than the motion region, additional noise will be included, which may degrade the displacement measurement performance. Therefore, an optimal center frequency exists such that the corresponding focus region matches the motion region well without introducing unnecessary noise.

$$\text{motion region} = \text{motion range} + \text{target size} \quad (14)$$



**Fig. 4.** Geometric relationship among the motion region, motion range, and target size.

The existence of an optimal center frequency is illustrated using the parameter combination with a displacement amplitude ratio of 0.040 and a target size ratio of 0.012, for which the optimal center frequency is 0.0100. As shown in Fig. 5, when the center frequency is lower than the optimal value, the MAE increases more gradually as the center frequency decreases, which can be attributed to the additional noise introduced by the larger focus region associated with a lower center frequency. In contrast, when the center frequency is higher than the optimal value, the MAE rises sharply, indicating a failure in displacement measurement. This is because the smaller focus region associated with a higher center frequency cannot fully cover the motion region. This phenomenon is further illustrated by the time history displacement measurement results shown in Fig. 6. During the initial stage of motion, when the displacement amplitude is large, the estimated displacement exhibits significant distortion because the focus region is too small to cover the entire motion region. In contrast, during the later stage, when the motion amplitude becomes smaller, the estimated motion pattern remains relatively accurate, since the smaller focus region is still sufficient to cover the low-amplitude portion of the vibration.

Based on the previous analysis, we speculate that the focus region of the Gabor wavelet needs to match the motion region of the target, as shown in Eq. (15), in order to achieve optimal measurement performance. Based on this speculation, we calculated the estimated optimal center frequencies for all numerical videos from the motion region according to Eq. (15). The estimated results and comparisons are presented in Table 2.

Estimated optimal center frequency

$$= \frac{1}{\text{focus region} = \text{motion region} = \text{motion range} + \text{target size}} \quad (15)$$

As observed in Table 2, although the estimated optimal center frequency is not exactly identical to the true optimal center frequency because the spacing between adjacent tested center frequencies is relatively large, it remains close to the true optimal value, and the two follow similar trends. These results support our speculation that the focus region of the Gabor wavelet needs to fit the motion region in order to achieve optimal measurement performance.

Therefore, a key equation is proposed in this section, as shown in Eq. (16). This equation describes the mathematical relationship between the optimal center frequency and the motion region of the target, and serves as the foundation for the proposed center frequency optimization method. It is worth noting that this relationship is defined entirely in pixel space. Therefore, Eq. (16) is independent of the imaging resolution

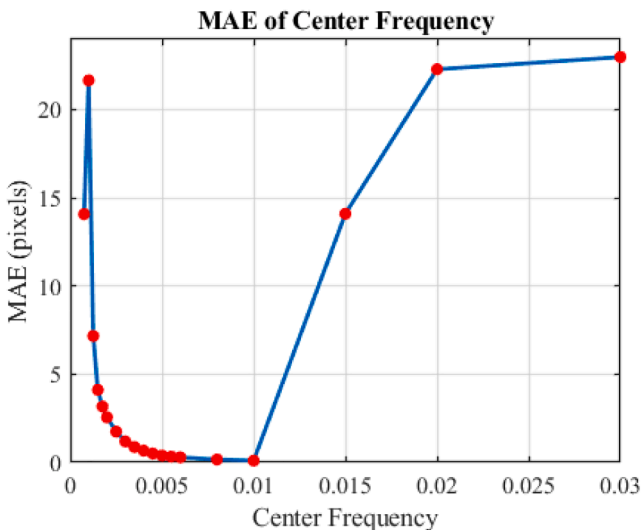


Fig. 5. The measurement accuracy (MAE) under different center frequencies.

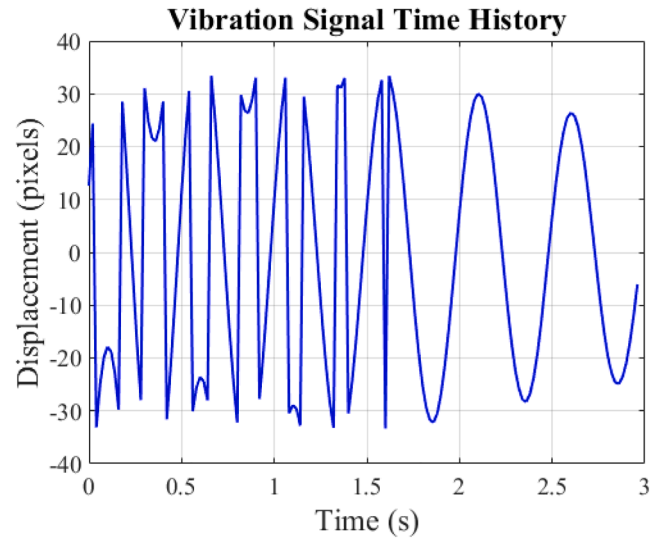


Fig. 6. Measured displacement time history at a center frequency of 0.03.

and applicable to videos of any resolution.

$$\text{optimal center frequency} = \frac{1}{\text{motion region}} \quad (16)$$

To provide further insight into why this inverse relationship holds, an analytical explanation is given as follows. In general, the focus region of the Gabor wavelet should match the motion region of the target to achieve optimal measurement performance. This argument is supported by two aspects. First, as illustrated in Fig. 7, when the Gabor wavelet possesses a larger focus region, it captures extraneous noise beyond the motion region, leading to a degradation in measurement accuracy. This is corroborated by the cases in Fig. 5, where it can be observed that when the center frequency is lower than the optimal center frequency (i.e., when the focus region exceeds the motion region), the measurement accuracy decreases gradually. This gradual degradation is attributed to the inclusion of noise from beyond the motion region. Second, as illustrated in Fig. 7, when the Gabor wavelet possesses a smaller focus region, it fails to capture the target motion that extends beyond the focus region, resulting in complete measurement failure. This is further verified in Fig. 6, where a displacement time history obtained with a center frequency higher than the optimal value (i.e., a focus region smaller than the motion region) shows that the measurement completely fails during the high-amplitude intervals, while it remains effective during the low-amplitude intervals. In summary, based on these two aspects, there exists an optimal focus region that matches the motion region to achieve the best displacement measurement performance. According to Eq. (13), the focus region is theoretically inversely proportional to the center frequency. Therefore, the optimal center frequency should be inversely proportional to the motion region, which is the relationship described by Eq. (16) and serves as the theoretical foundation of the proposed optimization method.

### 3.3. The optimization method for center frequency in Gabor wavelet for phase-based displacement measurement

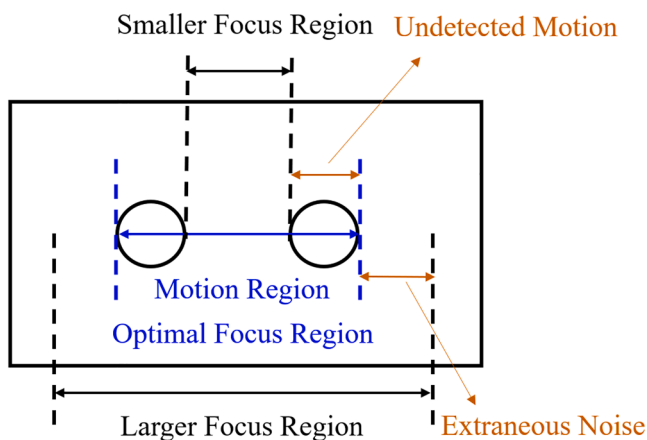
Based on the mathematical relationship between optimal center frequency and motion region proposed in the previous section, the center frequency optimization method is presented in this section, as illustrated in the following steps and Fig. 8:

- (1) Initialization of the center frequency: According to the video frame size, an initially sufficiently low center frequency is selected so that the entire possible motion region can be covered.

**Table 2**

Comparison of the estimated optimal center frequency and the optimal center frequency determined from ground truth data.

| Motion amplitude ratio | Target size ratio | Motion region (pixels) | Estimated optimal center frequency | Optimal center frequency |
|------------------------|-------------------|------------------------|------------------------------------|--------------------------|
| 0.004                  | 0.012             | 24.96                  | 0.04006                            | 0.05000                  |
| 0.004                  | 0.024             | 40.32                  | 0.02480                            | 0.02000                  |
| 0.004                  | 0.048             | 71.04                  | 0.01408                            | 0.01500                  |
| 0.004                  | 0.096             | 132.48                 | 0.00755                            | 0.00600                  |
| 0.004                  | 0.120             | 163.20                 | 0.00613                            | 0.00500                  |
| 0.004                  | 0.240             | 316.80                 | 0.00316                            | 0.00300                  |
| 0.008                  | 0.012             | 34.56                  | 0.02894                            | 0.03000                  |
| 0.008                  | 0.024             | 49.92                  | 0.02003                            | 0.01500                  |
| 0.008                  | 0.048             | 80.64                  | 0.01240                            | 0.01000                  |
| 0.008                  | 0.096             | 142.08                 | 0.00704                            | 0.00600                  |
| 0.008                  | 0.120             | 172.80                 | 0.00579                            | 0.00400                  |
| 0.008                  | 0.240             | 326.40                 | 0.00306                            | 0.00300                  |
| 0.040                  | 0.012             | 111.36                 | 0.00898                            | 0.01000                  |
| 0.040                  | 0.024             | 126.72                 | 0.00789                            | 0.00800                  |
| 0.040                  | 0.048             | 157.44                 | 0.00635                            | 0.00600                  |
| 0.040                  | 0.096             | 218.88                 | 0.00457                            | 0.00500                  |
| 0.040                  | 0.120             | 249.60                 | 0.00401                            | 0.00400                  |
| 0.040                  | 0.240             | 403.20                 | 0.00248                            | 0.00250                  |
| 0.080                  | 0.012             | 207.36                 | 0.00482                            | 0.00500                  |
| 0.080                  | 0.024             | 222.72                 | 0.00449                            | 0.00500                  |
| 0.080                  | 0.048             | 253.44                 | 0.00395                            | 0.00500                  |
| 0.080                  | 0.096             | 314.88                 | 0.00318                            | 0.00450                  |
| 0.080                  | 0.120             | 345.60                 | 0.00289                            | 0.00300                  |
| 0.080                  | 0.240             | 499.20                 | 0.00200                            | 0.00200                  |
| 0.100                  | 0.012             | 256.00                 | 0.00391                            | 0.00400                  |
| 0.100                  | 0.024             | 271.36                 | 0.00369                            | 0.00400                  |
| 0.100                  | 0.048             | 302.08                 | 0.00331                            | 0.00400                  |
| 0.100                  | 0.096             | 363.52                 | 0.00275                            | 0.00400                  |
| 0.100                  | 0.120             | 394.24                 | 0.00254                            | 0.00300                  |
| 0.100                  | 0.240             | 547.84                 | 0.00183                            | 0.00175                  |
| 0.150                  | 0.012             | 376.32                 | 0.00266                            | 0.00250                  |
| 0.150                  | 0.024             | 391.68                 | 0.00255                            | 0.00250                  |
| 0.150                  | 0.048             | 422.40                 | 0.00237                            | 0.00250                  |
| 0.150                  | 0.096             | 483.84                 | 0.00207                            | 0.00250                  |
| 0.150                  | 0.120             | 514.56                 | 0.00194                            | 0.00250                  |
| 0.150                  | 0.240             | 668.16                 | 0.00150                            | 0.00175                  |
| 0.200                  | 0.012             | 495.36                 | 0.00202                            | 0.00200                  |
| 0.200                  | 0.024             | 510.72                 | 0.00196                            | 0.00200                  |
| 0.200                  | 0.048             | 541.44                 | 0.00185                            | 0.00200                  |
| 0.200                  | 0.096             | 602.88                 | 0.00166                            | 0.00175                  |
| 0.200                  | 0.120             | 633.60                 | 0.00158                            | 0.00175                  |
| 0.200                  | 0.240             | 787.20                 | 0.00127                            | 0.00175                  |
| 0.250                  | 0.012             | 615.68                 | 0.00162                            | 0.00150                  |
| 0.250                  | 0.024             | 631.04                 | 0.00158                            | 0.00150                  |
| 0.250                  | 0.048             | 661.76                 | 0.00151                            | 0.00150                  |
| 0.250                  | 0.096             | 723.20                 | 0.00138                            | 0.00150                  |
| 0.250                  | 0.120             | 753.92                 | 0.00133                            | 0.00150                  |
| 0.250                  | 0.240             | 907.52                 | 0.00110                            | 0.00150                  |



**Fig. 7.** Illustration of the relationship between focus region and motion region for displacement measurement.

Based on this initial center frequency, the motion range is preliminarily estimated. Here, the motion range is defined as the maximum displacement variation over the entire motion period, i.e., the difference between the maximum and minimum displacements.

- (2) Estimation of the motion region: The target size is measured in the frame as the projected length of the target along the motion direction. The motion region is then estimated by combining the target size with the motion range.
- (3) Update of the center frequency: Based on the estimated motion region, a new center frequency is calculated according to Eq. (16) in Section 3.2.
- (4) Refinement of displacement and motion range: Using the updated center frequency, the displacement is re-measured and the motion range is re-estimated, which yields an updated motion region.
- (5) Recalculation of the center frequency: The center frequency is recalculated from the updated motion region according to Eq. (16) in Section 3.2.
- (6) Iteration until convergence: Steps 4 and 5 are repeated until the difference between two consecutive center frequency values is smaller than a predefined threshold. In this study, convergence is considered to be achieved when the two center frequency values are identical to three significant digits. The converged value is regarded as the optimal center frequency, which provides the most reliable displacement estimation.

In the proposed iterative optimization method, a bandwidth of 1 is adopted following prior studies (Fleet & Jepson, 1990; Liu et al., 2022), as discussed in Section 2.3. This choice balances the spatial and frequency localization capabilities of the Gabor wavelet. However, it is worth discussing whether the proposed center frequency selection rule is expected to remain valid when a different bandwidth is adopted.

Before discussing the effect of different bandwidth choices, it is important to note that the frequency domain response of the Gabor wavelet is a Gaussian-shaped function centered at the center frequency, with the amplitude gradually decaying toward both sides. When the bandwidth is increased, the frequency domain response of the Gabor wavelet broadens around the center frequency. However, the position of the center frequency remains unchanged, as it is still the peak frequency in the frequency domain response of the Gabor wavelet. Therefore, similar to the bandwidth of 1 adopted in this study, the peak frequency and its corresponding focus region can still be used to match the motion region, thereby determining the optimal center frequency. Meanwhile, since the Gabor wavelet inherently possesses a certain bandwidth, some frequencies around the center frequency still maintain a sufficiently high response. This means that even when the center frequency deviates slightly from the optimal value, the Gabor wavelet can still capture sufficient frequency components related to the motion region, providing measurement accuracy that, while slightly lower than the optimal, remains relatively high. Therefore, when the bandwidth is further increased, this tolerance becomes more pronounced, and the range of center frequencies around the optimal value that can provide relatively high measurement accuracy will broaden compared to the bandwidth of 1. However, the broader frequency domain response also introduces more interference from frequency components unrelated to the motion region. Moreover, a larger bandwidth is unfavorable for the Hilbert transform. Both factors may lead to a decrease in overall measurement accuracy.

Conversely, when the bandwidth is decreased, the frequency domain response of Gabor wavelet narrows around the center frequency. The position of the center frequency remains unaffected by the reduction in bandwidth, as it is still the peak frequency in the frequency domain response. Therefore, similar to the bandwidth of 1 adopted in this study, the focus region corresponding to the peak frequency can still be matched with the motion region to determine the optimal center

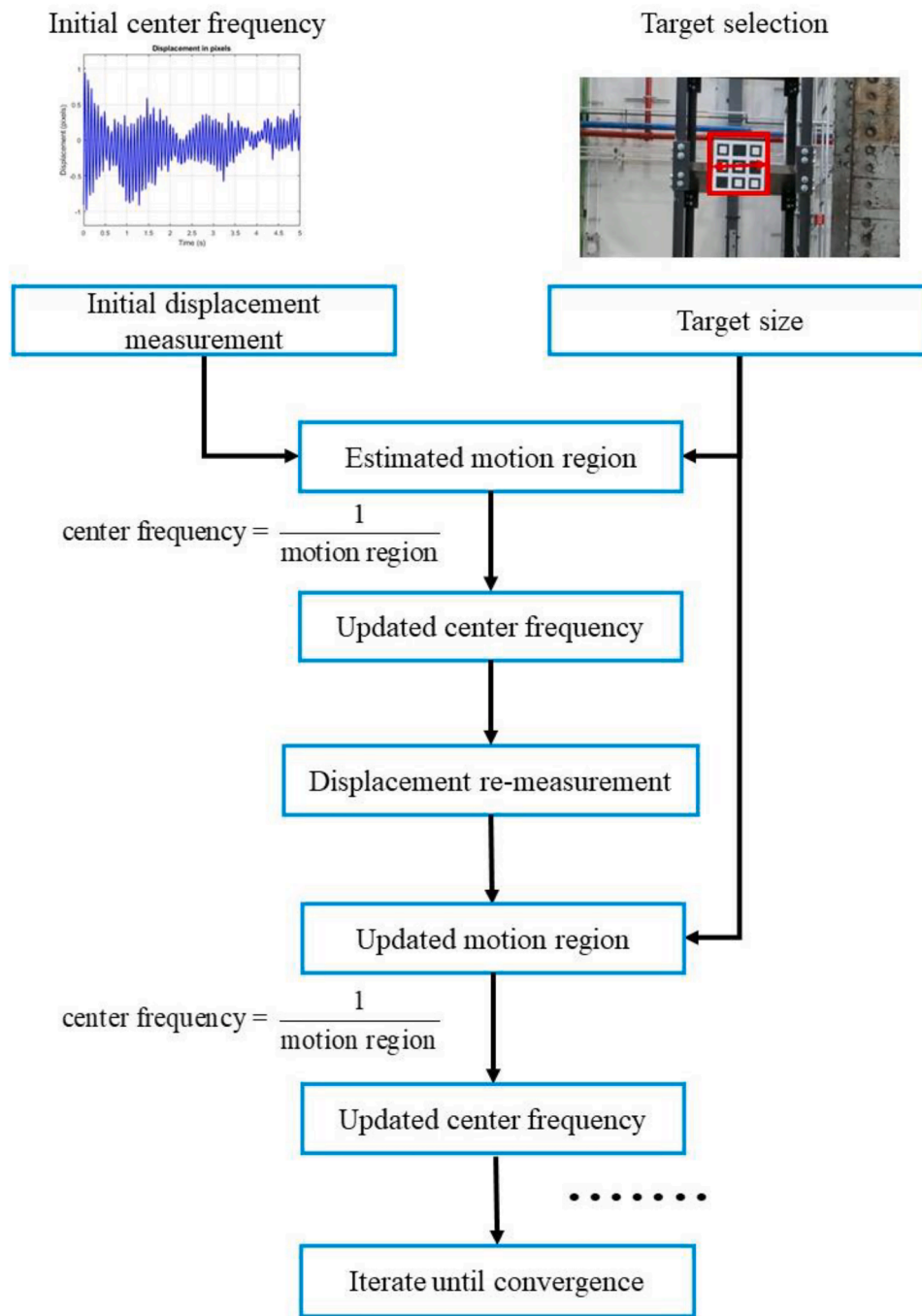


Fig. 8. Flowchart of the proposed center frequency optimization method.

frequency. Compared to the bandwidth of 1, some frequencies slightly deviating from the optimal center frequency can still achieve relatively near-optimal measurement accuracy, but the range of such frequencies will be further reduced. It should be noted that, although a smaller bandwidth is beneficial for isolating interference from frequency components unrelated to the motion region and is favorable for the Hilbert transform, it also corresponds to a larger Gabor wavelet in the spatial domain. This results in reduced target localization capability, which may ultimately affect the overall measurement accuracy.

#### 4. Experimental verification

##### 4.1. Verification experiment on a column fixed on a shaker

To validate the proposed optimization method for Gabor wavelet parameter selection in phase-based displacement measurement, a laboratory vision-based displacement test was performed. The test specimen was a column mounted on the end of a shaker and driven to undergo one-dimensional reciprocating motion. Video data were captured using an iPhone 11 at a frame rate of 24 fps and a resolution of 720P, whereas the corresponding displacement response was simultaneously measured by a Keyence LK-G3000 series laser displacement

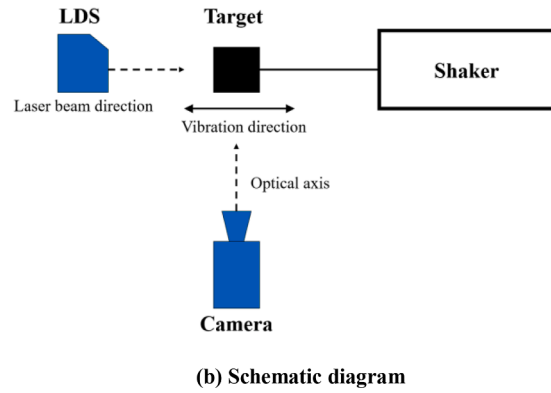
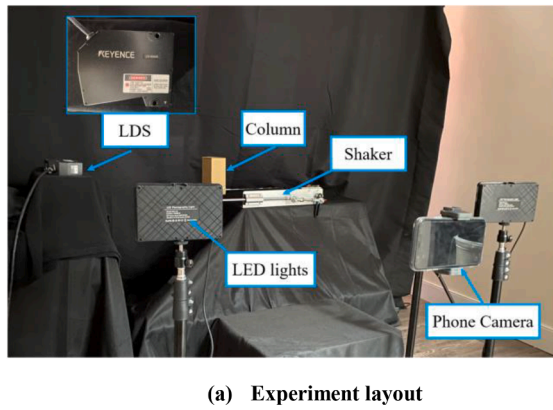


Fig. 9. Images of the experiment layout and schematic diagram.

sensor (LDS) for reference. Fig. 9(a) and (b) present the experimental layout and the associated schematic diagram, respectively.

The camera was arranged with its optical axis perpendicular to the motion center direction of the vibrating column. Throughout the experiment, the distance between the camera and the target was kept at 900 mm. A focal length of 26 mm yielded an image scale of approximately 0.81 mm/pixel. As indicated by the first recorded frame in Fig. 10, the column was initially positioned close to the center of the field of view.

Table 3 presents the optimization procedure based on the proposed parameter optimization method. At the beginning of the optimization, a low center frequency was selected to obtain a large focus region that could cover all possible motion regions in the frame. Since the target moves along the width of the 720P frame, which is 1280 pixels, the center frequency was set to 0.001, corresponding to a focus region of 1000 pixels. The frequency domain response of the Gabor filter is not a strict cutoff function but a band-pass response with gradual attenuation around the center frequency. As a result, motion regions with scales slightly beyond the 1000-pixel range may still be effectively captured when the center frequency is 0.001.

Based on the large focus region corresponding to the initial center frequency, the displacement measurement results are first obtained. The motion range is then determined from the displacement results to be 43.6324 pixels, which corresponds to the maximum motion range during the entire vibration period. The target size is measured as the width of the target along the motion direction in the first frame and is found to be 65 pixels. Based on the motion range and target size, the motion region is determined to be 108.6324 pixels. The corresponding frequency of this motion region is 0.00921, which is used as the estimated center frequency for the next iteration. It should be noted that this estimated center frequency differs significantly from the initial value of 0.001, indicating that further iterations are necessary to identify the optimal center frequency.

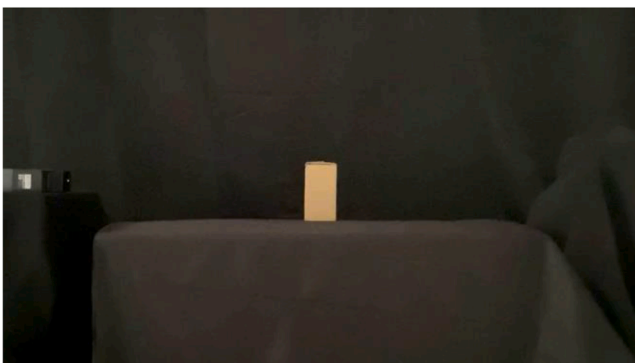


Fig. 10. The first frame of the recorded motion video.

Table 3

Iterative center frequency optimization process of the proposed method in the laboratory shaker test.

|                                      | target size (pixels) | motion range (pixels) | motion region (pixels) | estimated center frequency | MAE (mm) |
|--------------------------------------|----------------------|-----------------------|------------------------|----------------------------|----------|
| center frequency = 0.001             | 65                   | 43.6324               | 108.6324               | 0.00921                    | 3.1507   |
| center frequency = 0.00921           | 65                   | 39.7113               | 104.7113               | 0.00955                    | 0.3204   |
| center frequency = 0.00955 (optimal) | 65                   | 39.7431               | 104.7431               | 0.00955                    | 0.3174   |

Then, displacement measurements are performed using a center frequency of 0.00921, and the motion range is determined to be 39.7113 pixels. Similarly, based on the target size and motion range, the motion region is calculated to be 104.7113 pixels. Based on this motion region, the estimated center frequency in the second iteration is 0.00955. Following the same procedure, a third iteration is performed. The estimated center frequency obtained in the third iteration is also 0.00955, which is identical to that obtained in the previous iteration. This indicates that the proposed parameter optimization method has converged and that stable displacement measurement results have been achieved.

The convergence process is further visualized in Fig. 11, which presents the center frequency and MAE at each iteration step. As shown, the input center frequency increases rapidly from the initial value of 0.001 toward the estimated center frequency, and the two lines converge at the second iteration step, indicating that the optimization has reached a stable state. Meanwhile, the MAE decreases sharply from 3.1507 mm at the initial step to 0.3204 mm at the first iteration, and further stabilizes at 0.3174 mm at the second iteration. This demonstrates that the proposed method achieves rapid convergence, with the center frequency and measurement accuracy both stabilizing within only two iteration steps.

The time history displacement measurement results obtained using the Gabor wavelet method with the optimal center frequency are presented in Fig. 12. As can be observed, the optimal center frequency provides accurate displacement measurements of the column motion over the entire period.

To verify whether the optimal center frequency determined by the proposed optimization method is valid, the MAE values at each iteration step are also presented in Table 3. As shown, the MAE value decreases continuously as the number of iteration steps increases. In the converged iteration step, the MAE reaches a reasonably low value of 0.3174 mm,

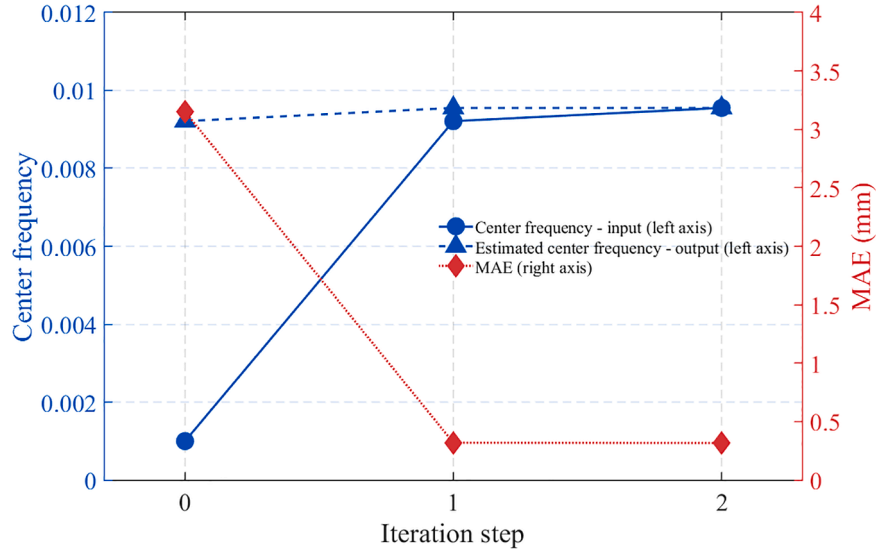


Fig. 11. Convergence plot of the iterative center frequency optimization process in the laboratory shaker test.

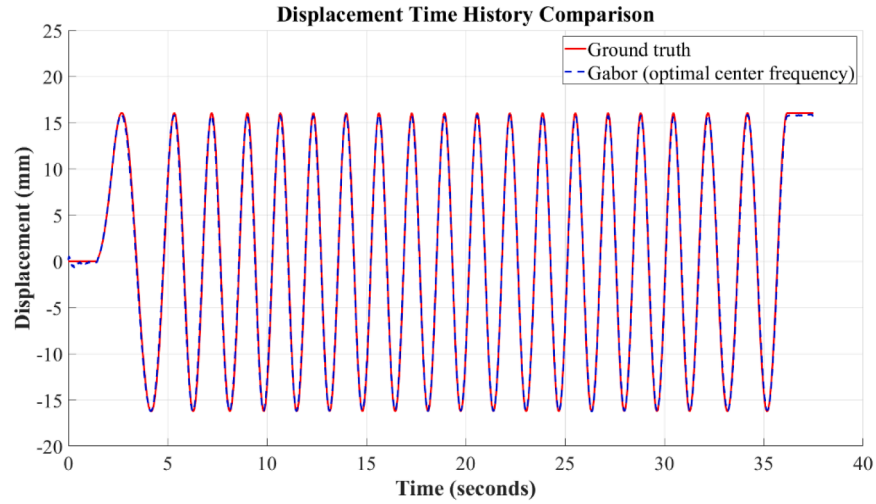


Fig. 12. Comparison of displacement history between Gabor wavelet (optimal center frequency) and laser measurements.

indicating accurate displacement measurement performance for this task.

Furthermore, to rule out the possibility that the proposed method is trapped in a local optimum, we additionally used other candidate center frequencies for displacement measurement, and the corresponding results are reported in Table 4. As shown in Table 4, the MAEs of these cases are consistently much higher than the MAE obtained with the optimal center frequency, further highlighting the importance of center frequency selection for displacement measurement accuracy and confirming that the optimal center frequency determined by the proposed

Table 4

The measurement performance of the Gabor wavelet method with other candidate center frequencies.

| Center frequency | MAE (mm) |
|------------------|----------|
| 0.0020           | 2.7517   |
| 0.0140           | 1.0936   |
| 0.0500           | 8.4591   |
| 0.1000           | 10.0541  |
| 0.2000           | 10.0545  |
| 0.5000           | 10.0541  |

optimization method indeed yields the best performance.

To further evaluate the displacement measurement performance of the Gabor wavelet method with the optimal center frequency, several other methods, including phase-based optical flow based on Complex Steerable Pyramid (CSP), LK optical flow, template matching, and deep learning-based optical flow (RAFT), are adopted for comparison. The parameter choices and region of interest (ROI) selection for each benchmark method are as follows. For the CSP method, the scale level is the key parameter, which plays a similar role to the center frequency in the Gabor wavelet method, as it determines whether the focused scale matches the motion region of the target (Ai et al., 2025). In this study, the optimal scale level was determined through comparison with the ground truth displacement data, and the corresponding displacement measurement result was used for the benchmark comparison. For the LK optical flow method, the selection of the tracking point is critical, as the method requires sufficient feature information to reliably track the target. The tracking point was selected on the edge of the target perpendicular to the motion direction, where a pronounced pixel intensity gradient aligned with the motion direction is present, providing a strong feature for motion estimation (Lucas & Kanade, 1981). For the template matching method, the key considerations are the selection of

the template and the matching criterion. The template, or ROI, was selected to encompass the entire target. The Normalized Cross-Correlation Coefficient was adopted as the matching criterion, as it eliminates the influence of absolute brightness differences between the template and the search region through normalization, providing robust matching stability. For the RAFT method, the pre-trained raft-things model was adopted, which has been recently employed for structural displacement measurement and demonstrated good sub-pixel measurement accuracy (Guo et al., 2025). The number of inference iterations was set to 20, following the default setting recommended in Teed and Deng (2020). The time history displacement measurement results are compared in Fig. 13, and the MAE of each method is reported in Table 5. As shown, the Gabor wavelet method with the optimal center frequency achieves the highest measurement accuracy among all the compared methods, further confirming the validity of the optimal center frequency determined by the proposed optimization method.

#### 4.2. The measurement of seismic response of the cold-formed steel wall (CSW) structure

To further validate the effectiveness of the proposed center-frequency optimization method for the Gabor wavelet in phase-based displacement measurement under real-world conditions, an experimental dataset from a large outdoor shaking-table test of a cold-formed steel wall (CSW) system was adopted. The data were obtained from the publicly accessible DesignSafe repository, and the corresponding tests were conducted at the Large High Performance Outdoor Shake Table (LHPOST) at the University of California, San Diego (UCSD) (Singh et al., 2021).

In the test setup, the walls were arranged in a shear-gravity–gravity–shear layout. Tie rods were provided at the ends of the shear-wall segments, and the steel sheathing was fastened according to the prescribed detailing scheme adopted for realistic construction simulation. Further information on the specimen design, instrumentation, and experimental records is available in (Singh et al., 2021).

As illustrated in Fig. 14, the structural response was recorded simultaneously by contact sensors and a video-based monitoring system. The table was driven by the CNP196 component of the Canoga Park record from the 1994 Northridge earthquake ( $M_w = 6.7$ ). A DVR camera was positioned normal to the excitation axis to capture the overall motion of the specimen, and the recorded images had a resolution of  $1920 \times 1080$  pixels at 30 fps. According to the calibration reported by Cai et al. (Cai et al., 2024), the pixel-to-length conversion ratio was 4.4418 mm/pixel.

To facilitate vision-based displacement monitoring, a marker was

**Table 5**

The vision-based methods measurement performances for the experiment case.

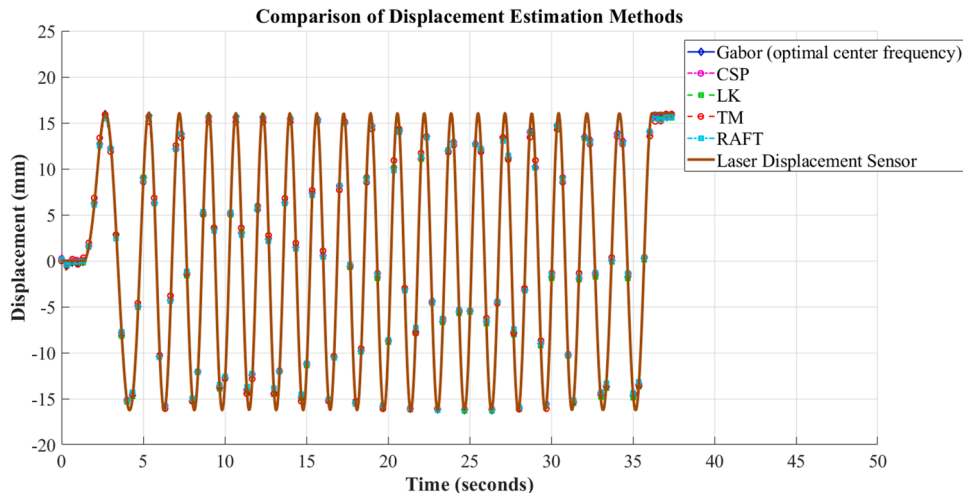
|          | Gabor  | CSP    | LK     | TM     | RAFT   |
|----------|--------|--------|--------|--------|--------|
| MAE (mm) | 0.3174 | 0.3457 | 0.3649 | 0.4681 | 0.4034 |

attached near the top region of the CSW specimen. Meanwhile, string potentiometers installed on the transfer beam were used to record the lateral displacement response for comparison and validation. These sensors operated at a sampling rate of 256 Hz, and their measurements were taken as the ground truth data for evaluating the vision-based method. It is worth noting that the background surrounding the marker in this experiment contains various structural components such as steel framing members, steel studs, connections, ground-mounted equipment, and the distant natural environment, representing a typical nonuniform and cluttered background commonly encountered in real-world structural engineering monitoring.

Using the proposed center frequency optimization method, a low center frequency is first selected as the initial center frequency to cover all possible motion regions in the video. The initial center frequency is set to 0.00067, which is lower than that used in the previous section and thus corresponds to a larger focus region. This is because the video has a resolution of 1080P, with 1920 pixels along the motion direction of the structure. The target is chosen as the marker at the top of the frame structure, as shown in Fig. 14. The target size along the motion direction is measured to be 91 pixels. When the center frequency is set to 0.00067, the motion range is determined to be 95.3114 pixels. Based on the target size and motion range, the motion region is determined to be 186.3114 pixels. Accordingly, the estimated center frequency derived from this motion region is 0.00537 in the initial iteration step.

Based on the displacement measurement results obtained with a center frequency of 0.00537, the same procedure as that in the initial iteration step is repeated, and the estimated center frequency in the second iteration step is determined to be 0.00630. An obvious difference is observed between the estimated center frequencies in the initial and second iteration steps, indicating that further iterations are required. Therefore, additional iteration steps are performed, and the final optimal center frequency is obtained as 0.00628 at the fourth iteration step, as shown in Table 6.

The convergence process is further visualized in Fig. 15. As shown, the input center frequency increases from the initial value of 0.00067 and progressively approaches the estimated center frequency over the iteration steps. The two lines converge at the third iteration step, indicating that the optimization has reached a stable state. The MAE decreases significantly from 7.1190 mm at the initial step to 1.1643 mm at



**Fig. 13.** Comparison of displacement history between vision-based methods and laser measurements.

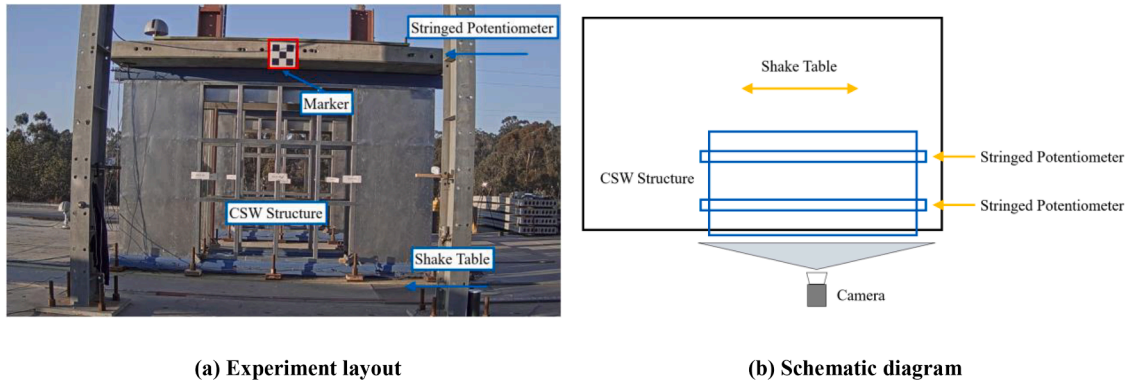


Fig. 14. Experimental layout of the shaking table test of the CSW structure (adapted from Singh et al. (2021)).

Table 6

Iterative center frequency optimization process of the proposed method in the shaking table test of the CSW structure.

|                                      | target size (pixels) | motion range (pixels) | motion region (pixels) | estimated center frequency | MAE (mm) |
|--------------------------------------|----------------------|-----------------------|------------------------|----------------------------|----------|
| center frequency = 0.00067           | 91                   | 95.3114               | 186.3114               | 0.00537                    | 7.1190   |
| center frequency = 0.00537           | 91                   | 67.6115               | 158.6115               | 0.00630                    | 1.1643   |
| center frequency = 0.00630           | 91                   | 68.2835               | 159.2835               | 0.00628                    | 1.0909   |
| center frequency = 0.00628 (optimal) | 91                   | 68.2698               | 159.2698               | 0.00628                    | 1.0885   |

the first iteration, and further stabilizes at 1.0885 mm at the third iteration. This confirms that the proposed method also achieves effective convergence under real-world conditions with more complex backgrounds.

The time history displacement measurement results obtained with the optimal center frequency are presented in Fig. 16. As shown, even for a random displacement signal containing multiple frequency components, the proposed optimization method remains effective and provides reliable measurements.

It should be noted that more iteration steps are required in this task than in the previous section. This is because the background is much more complex in this real-world case. As a result, the large focus region corresponding to the initial center frequency may contain more noise, which can affect the measurement accuracy and lead to a less accurate estimation of the motion range. This, in turn, may result in less accurate estimates of the motion region and center frequency for the next

iteration step. Therefore, more iteration steps are needed to achieve convergence.

To verify the validity of the optimal center frequency determined by the proposed optimization method, the MAE values at each iteration step are presented in Table 6. As shown, the MAE decreases progressively as the number of iteration steps increases and reaches a significantly low value of 1.0885 mm at the optimal center frequency. The successful application of the proposed method to 1080P video in this case confirms that it is not restricted to the 720P resolution used in the numerical experiments. This is because the proposed method is fundamentally built upon the relationship in Eq. (16), which is defined in pixel space and is inherently resolution-independent.

It is also worth noting that several structural components surrounding the marker, including the transfer beam, steel columns, steel studs, and the GPS antenna mounted on top of the specimen, all undergo motion during seismic excitation, constituting multiple moving regions

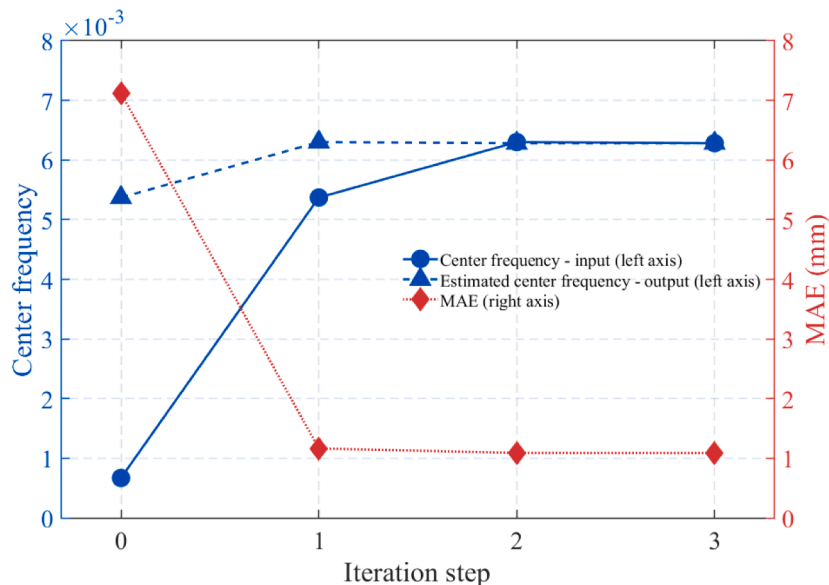


Fig. 15. Convergence plot of the iterative center frequency optimization process in the shaking table test of the CSW structure.

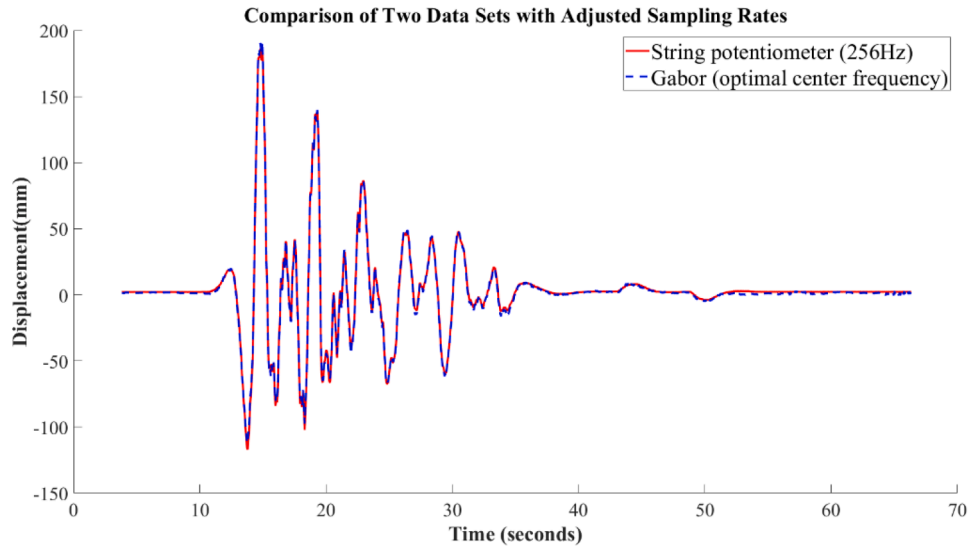


Fig. 16. Comparison of displacement history between Gabor wavelet (optimal center frequency) and string potentiometer measurements.

Table 7

The measurement performance of the Gabor wavelet method with other candidate center frequencies.

| Center frequency | MAE (mm) |
|------------------|----------|
| 0.0040           | 3.4354   |
| 0.0090           | 3.9013   |
| 0.0120           | 14.9528  |
| 0.0500           | 12.9843  |
| 0.2000           | 15.6047  |
| 0.5000           | 15.6037  |

does not introduce significant interference to the measurement of the target marker.

Furthermore, to rule out the possibility that the optimal center frequency corresponds to a local optimum, the MAE results obtained using other candidate center frequencies that may appear in the frame are also calculated for displacement measurement and presented in Table 7. As shown, the MAE values of these cases are much higher than that obtained with the optimal center frequency, further verifying the validity of the optimal center frequency and highlighting the importance of center frequency selection in real-world cases with more complex backgrounds.

To further demonstrate the effectiveness of the optimal center frequency, several other methods employed in the previous section are employed for comparison in this measurement task, as listed in Table 8. As shown, the Gabor wavelet method with the optimal center frequency achieves the highest measurement accuracy among all the compared phase-based, intensity-based, and deep learning-based methods. The time history displacement measurement comparisons of the different methods are presented in Fig. 17.

Table 8

MAE comparison between different vision-based methods for the shaking table test of the CSW structure.

|          | Gabor  | CSP    | LK     | TM     | RAFT   |
|----------|--------|--------|--------|--------|--------|
| MAE (mm) | 1.0885 | 1.3397 | 2.6180 | 2.8131 | 5.8719 |

within the field of view. Despite the presence of these adjacent moving regions, the proposed method still achieved accurate displacement measurement results, suggesting that the motion of nearby components

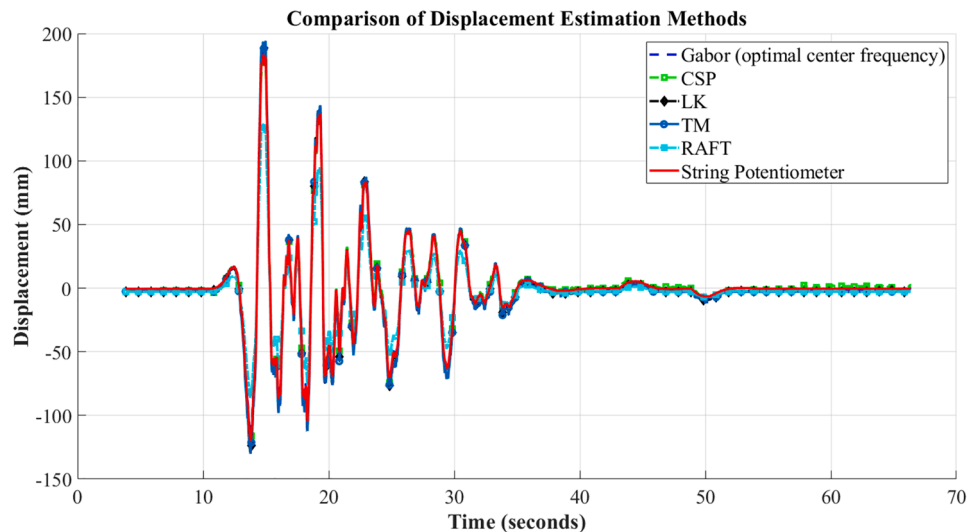


Fig. 17. Comparison of displacement history between vision-based methods and string potentiometer.

## 5. Conclusions

This study proposes a motion-region-based center frequency optimization method for Gabor wavelets in phase-based displacement measurement. Through numerical experiments on numerical videos with different motion ranges and target sizes, it is found that the optimal center frequency is not fixed, but varies systematically with the target motion characteristics. In particular, the numerical results reveal a clear relationship between the optimal center frequency and the motion region of the target, indicating that larger motion regions generally require lower center frequencies for accurate displacement measurement.

Based on this observation from the numerical experiments, a physics-inspired relationship between the optimal center frequency and the target motion region is established from the physical interpretation of the Gabor wavelet. On this basis, an iterative optimization method is developed to determine the center frequency directly from video data without relying on ground truth displacement information.

The proposed method is further validated through a laboratory shaker test and a seismic-response measurement of a cold-formed steel wall structure. The experimental results demonstrate that selecting the optimal center frequency significantly improves displacement measurement accuracy relative to other center frequency settings and enables the proposed method to achieve highly competitive accuracy compared with other vision-based displacement measurement methods under different testing conditions. In particular, the CSW shaking table test demonstrated that the proposed method remains effective under nonuniform and cluttered background conditions with the presence of multiple moving regions, which are typical in real-world structural engineering monitoring.

Overall, this study provides an effective and practical solution for center frequency selection for phase-based displacement measurement using Gabor wavelets, thereby improving its applicability in real-world structural vibration monitoring. However, the proposed method is primarily intended for scenarios involving a single target with dominant one-dimensional motion and a relatively ideal imaging configuration. Future work will focus on extending the proposed method to two-dimensional motion measurement, irregular target shapes, multiple targets with distinct motion characteristics, as well as varying imaging conditions.

## CRedit authorship contribution statement

**Sida Ai:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zhenkun Li:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Formal analysis.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

## References

- Ai, S., Dong, C., & Mei, Q. (2025). The optimal scale level of complex steerable pyramid for phase-based motion estimation under different motion ranges and target sizes. *Measurement*, 253, Article 117440. <https://doi.org/10.1016/j.measurement.2025.117440>
- Barron, J. L., Fleet, D. J., Beauchemin, S. S., & Burkitt, T. A. (1992). Performance of optical flow techniques. In *Proceedings 1992 IEEE Computer Society Conference on*

- Computer Vision and Pattern Recognition* (pp. 236–242). IEEE Comput. Soc. Press. <https://doi.org/10.1109/CVPR.1992.223269>.
- Cai, E., Zhang, Y., Lu, X., Ji, X., Gao, X., Hou, J., Shi, J., & Guo, W. (2024). Estimating structural motions in extreme environmental conditions—a dynamic correlation filter based computer vision approach. *Mechanical Systems and Signal Processing*, 215, Article 111398. <https://doi.org/10.1016/j.ymssp.2024.111398>
- Cai, E., Zhang, Y., & Quek, S. T. (2023). Visualizing and quantifying small and nonstationary structural motions in video measurement. *Computer-Aided Civil and Infrastructure Engineering*, 38, 135–159. <https://doi.org/10.1111/mice.12894>
- Chen, Y., Joffre, D., & Avitabile, P. (2018). Underwater dynamic response at limited points expanded to full-field strain response. *Journal of Vibration and Acoustics*, 140, Article 051016. <https://doi.org/10.1115/1.4039800>
- Collier, S., & Dare, T. (2022). Accuracy of phase-based optical flow for vibration extraction. *Journal of Sound and Vibration*, 535, Article 117112. <https://doi.org/10.1016/j.jsv.2022.117112>
- Daugman, J. G. (1985). Uncertainty relation for resolution in space, spatial frequency, and orientation optimized by two-dimensional visual cortical filters. *Journal of the Optical Society of America A*, 2, 1160. <https://doi.org/10.1364/JOSAA.2.001160>
- Diamond, D. H., Heyns, P. S., & Oberholster, A. J. (2017). Accuracy evaluation of sub-pixel structural vibration measurements through optical flow analysis of a video sequence. *Measurement*, 95, 166–172. <https://doi.org/10.1016/j.measurement.2016.10.021>
- Dong, C.-Z., & Catbas, F. N. (2021). A review of computer vision-based structural health monitoring at local and global levels. *Structural Health Monitoring*, 20, 692–743. <https://doi.org/10.1177/1475921720935585>
- Feng, D., & Feng, M. Q. (2018). Computer vision for SHM of civil infrastructure: From dynamic response measurement to damage detection – a review. *Engineering Structures*, 156, 105–117. <https://doi.org/10.1016/j.engstruct.2017.11.018>
- Fleet, D. J., & Jepson, A. D. (1990). Computation of component image velocity from local phase information. *International Journal of Computer Vision*, 5, 77–104. <https://doi.org/10.1007/BF00056772>
- Fogel, I., & Sagi, D. (1989). Gabor filters as texture discriminator. *Biological Cybernetics*, 61, 103–113. <https://doi.org/10.1007/BF00204594>
- Gautama, T., & Van Hulle, M. A. (2002). A phase-based approach to the estimation of the optical flow field using spatial filtering. *IEEE Transactions on Neural Networks*, 13, 1127–1136. <https://doi.org/10.1109/TNN.2002.1031944>
- Gomez, F., Park, J.-W., & Spencer, B. F. (2018). Reference-free structural dynamic displacement estimation method. *Structural Control & Health Monitoring*, 25, Article e2209. <https://doi.org/10.1002/stc.2209>
- Guo, L., Guo, W., Chen, D., Duan, B., & Shi, Z. (2025). Structural vibration measurement based on improved phase-based motion magnification and deep learning. *Mechanical Systems and Signal Processing*, 224, Article 111945. <https://doi.org/10.1016/j.ymssp.2024.111945>
- Horn, B. K. P., & Schunck, B. G. (1981). Determining optical flow. *Artificial Intelligence*, 17, 185–203. [https://doi.org/10.1016/0004-3702\(81\)90024-2](https://doi.org/10.1016/0004-3702(81)90024-2)
- Li, M., Liu, G., & Mao, Z. (2024). Phased-based motion estimation through short-distance hilbert transform. *Mechanical Systems and Signal Processing*, 211, Article 111219. <https://doi.org/10.1016/j.ymssp.2024.111219>
- Liu, G., Li, M. Z., Mao, Z., & Yang, Q. S. (2022). Structural motion estimation via hilbert transform enhanced phase-based video processing. *Mechanical Systems and Signal Processing*, 166, Article 108418. <https://doi.org/10.1016/j.ymssp.2021.108418>
- Liu, S., Yu, L., Niu, W., Wang, J., Zhong, Z., Huang, J., & Shan, M. (2024). Fast and accurate visual vibration measurement via derivative-enhanced phase-based optical flow. *Mechanical Systems and Signal Processing*, 209, Article 111089. <https://doi.org/10.1016/j.ymssp.2023.111089>
- Lucas, B. D., & Kanade, T. (1981). An iterative image registration technique with an application to stereo vision. In *ICAP'81: 7th International Joint Conference on Artificial Intelligence* (pp. 674–679). <https://hal.science/hal-03697340>.
- Miao, Y., Jeon, J. Y., Kong, Y., & Park, G. (2022). Phase-based displacement measurement on a straight edge using an optimal complex Gabor filter. *Mechanical Systems and Signal Processing*, 164, Article 108224. <https://doi.org/10.1016/j.ymssp.2021.108224>
- Miao, Y., Kong, Y., Nam, H., Lee, S., & Park, G. (2023a). Phase-based vibration imaging for structural dynamics applications: Marker-free full-field displacement measurements with confidence measures. *Mechanical Systems and Signal Processing*, 198, Article 110418. <https://doi.org/10.1016/j.ymssp.2023.110418>
- Miao, Y., Kong, Y., Young Jeon, J., Nam, H., & Park, G. (2023b). A novel marker for robust and accurate phase-based 2D motion estimation from noisy image data. *Mechanical Systems and Signal Processing*, 187, Article 109931. <https://doi.org/10.1016/j.ymssp.2022.109931>
- Morrone, M. C., Ross, J., Burr, D. C., & Owens, R. (1986). Mach bands are phase dependent. *Nature*, 324, 250–253. <https://doi.org/10.1038/324250a0>
- Movellan, J. R. (2002). Tutorial on Gabor filters. *Open Source Document*, 40, 1–23.
- Portilla, J., & Simoncelli, E. P. (2000). A parametric texture model based on joint statistics of complex wavelet coefficients. *International Journal of Computer Vision*, 40, 49–70. <https://doi.org/10.1023/A:1026553619983>
- Sarrafi, A., Mao, Z., Niezrecki, C., & Poozesh, P. (2018). Vibration-based damage detection in wind turbine blades using phase-based Motion Estimation and motion magnification. *Journal of Sound and Vibration*, 421, 300–318. <https://doi.org/10.1016/j.jsv.2018.01.050>
- Saxena, S. C., & Sekse, S. B. L. (1989). A self-compensated smart LVDT transducer. *IEEE Transactions on Instrumentation and Measurement*, 38, 748–753. <https://doi.org/10.1109/19.32186>
- Simpson, D. M. (1993). An introduction to the discrete orthogonal wavelet transform. *Revista Brasileira de Engenharia*, 9, 57–81.

- Singh, A., Hutchinson, T., Wang, X., Zhang, Z., Schafer, B., Derveni, F., ... Peterman, K. Wall line tests: Phase 1 – shake table tests. In CFS-NHERI: Seismic resiliency of repetitively framed mid-rise cold-formed steel buildings (Version 2). DesignSafe-CI. <https://doi.org/10.17603/DS2-MVJ8-8386>.
- Teed, Z., & Deng, J. (2020). RAFT: Recurrent all-pairs field transforms for optical flow. In A. Vedaldi, H. Bischof, T. Brox, & J.-M. Frahm (Eds.), *Computer vision – eccv 2020* (pp. 402–419). Cham: Springer International Publishing. [https://doi.org/10.1007/978-3-030-58536-5\\_24](https://doi.org/10.1007/978-3-030-58536-5_24).
- Wang, Y., Hu, W., Teng, J., & Xia, Y. (2023). Phase-based motion estimation in complex environments using the illumination-invariant log-Gabor filter. *Mechanical Systems and Signal Processing*, 186, Article 109847. <https://doi.org/10.1016/j.ymssp.2022.109847>
- Xu, Y., & Brownjohn, J. M. W. (2018). Review of machine-vision based methodologies for displacement measurement in civil structures. *Journal of Civil Structural Health Monitoring*, 8, 91–110. <https://doi.org/10.1007/s13349-017-0261-4>
- Yoon, H., Elanwar, H., Choi, H., Golparvar-Fard, M., & Spencer, B. F. (2016). Target-free approach for vision-based structural system identification using consumer-grade cameras: Target-free vision-based Structural System identification. *Structural Control and Health Monitoring*, 23, 1405–1416. <https://doi.org/10.1002/stc.1850>
- Zhang, D., Guo, J., Lei, X., & Zhu, C. (2016). A high-speed vision-based sensor for dynamic vibration analysis using fast motion extraction algorithms. *Sensors*, 16, 572. <https://doi.org/10.3390/s16040572>
- Zhang, W., Wang, X., Li, H., Zhou, J., Luo, Z., & Meng, G. (2024). Vibration measurement from an adaptive phase-based motion estimation using parameter optimised log-gabor filter. *Measurement*, 224, Article 113812. <https://doi.org/10.1016/j.measurement.2023.113812>